

## Program4: Integrate git with docker for source controlled application build

**Step1:** create a directory and install npm

mkdir program4

cd program4

To install npm

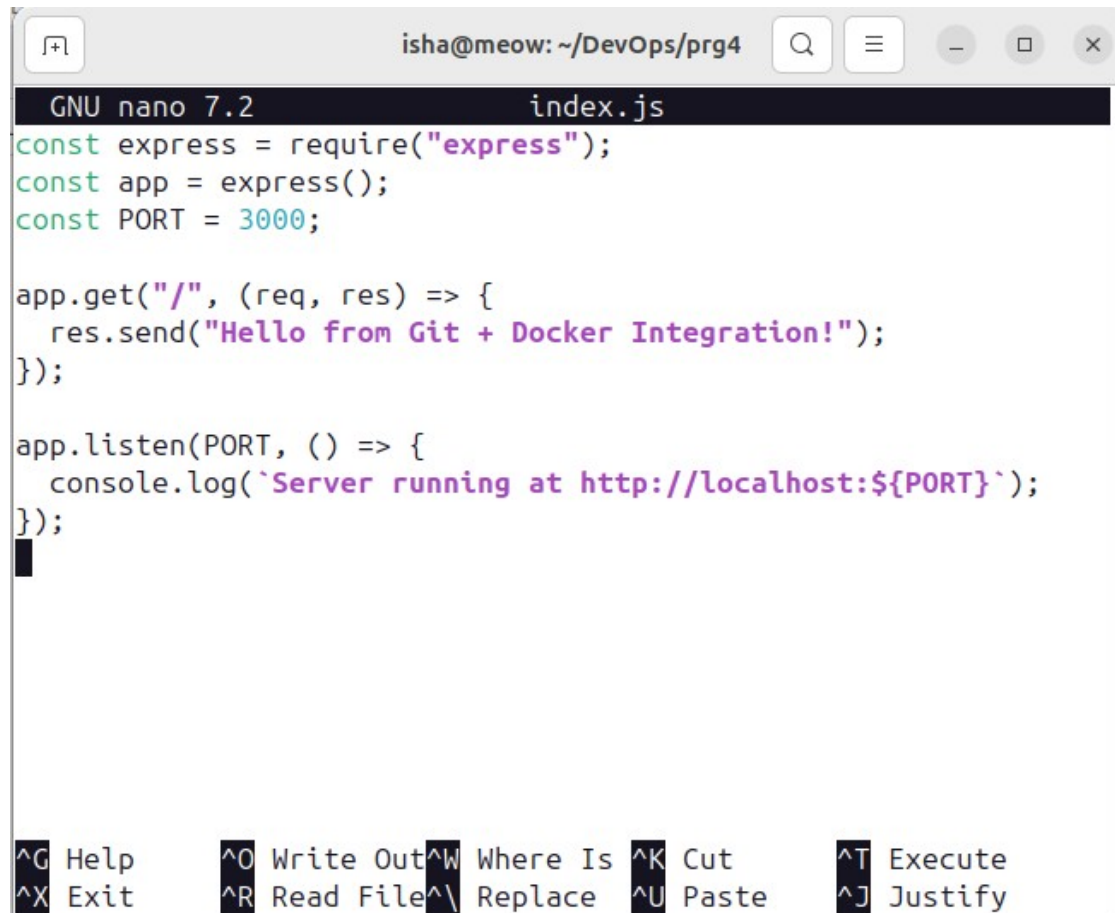
>> npm init -y

```
isha@meow:~/DevOps$ cd prg4
isha@meow:~/DevOps/prg4$ npm init -y
Wrote to /home/isha/DevOps/prg4/package.json:

{
  "name": "prg4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  },
  "devDependencies": {},
  "description": ""
}
```

```
isha@meow:~/DevOps/prg4$ █
```

**Step2:** create index.js and install express(npm install express)  
>> nano index.js



```
GNU nano 7.2 index.js
const express = require("express");
const app = express();
const PORT = 3000;

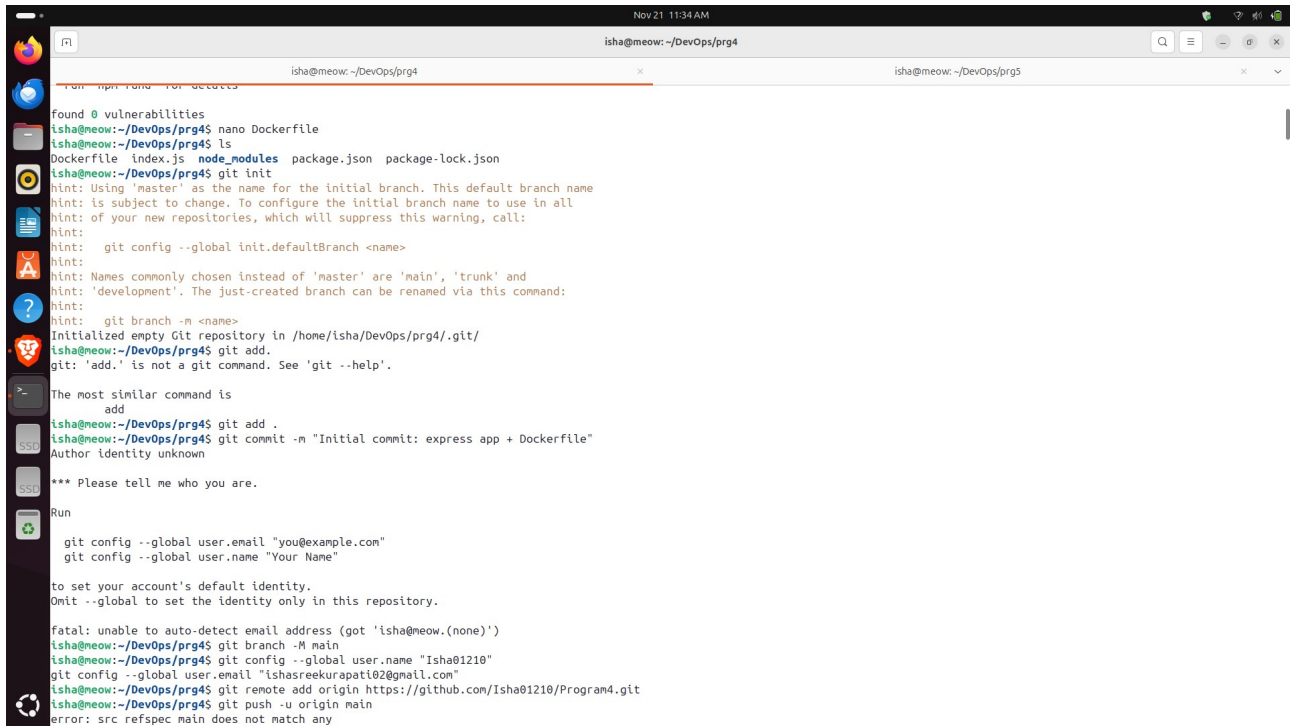
app.get("/", (req, res) => {
  res.send("Hello from Git + Docker Integration!");
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
isha@meow:~/DevOps/node-docker-app$ docker run -d -p 3000:3000 --name node-container node-docker-app
16cabd9e1e122afe28c0f7d4ba48b0ebb740e5cc04d98c02491b2c94a05319e4
isha@meow:~/DevOps/node-docker-app$
```

### Step3: create a repository in github and and commands to push the source code

A terminal window titled 'isha@meow: ~/DevOps/prg4' showing the execution of various git commands. The user has initialized a repository, added files, and attempted to push to a remote repository. The terminal output includes several hints from git, a fatal error about the email address, and a final error message indicating a mismatch between the local branch and the remote.

```
Nov 21 11:34 AM
isha@meow: ~/DevOps/prg4
Found 0 vulnerabilities
isha@meow:~/DevOps/prg4$ nano Dockerfile
isha@meow:~/DevOps/prg4$ ls
Dockerfile index.js node_modules package.json package-lock.json
isha@meow:~/DevOps/prg4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/isha/DevOps/prg4/.git/
isha@meow:~/DevOps/prg4$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
add
isha@meow:~/DevOps/prg4$ git add .
isha@meow:~/DevOps/prg4$ git commit -m "Initial commit: express app + Dockerfile"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

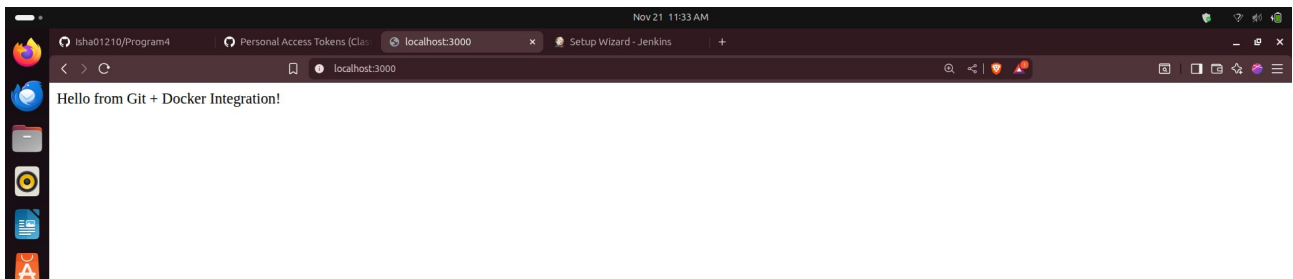
fatal: unable to auto-detect email address (got 'isha@meow.(none)')
isha@meow:~/DevOps/prg4$ git branch -M main
isha@meow:~/DevOps/prg4$ git config --global user.name "Isha01210"
git config --global user.email "ishasreekurapati02@gmail.com"
isha@meow:~/DevOps/prg4$ git remote add origin https://github.com/Isha01210/Program4.git
isha@meow:~/DevOps/prg4$ git push -u origin main
error: src refspec main does not match any
```

**Step4:** Build the Dockerfile and to see the output enter the url in the browser

```
ish@meow: ~/DevOps/node-docker-app
}

ish@meow:~/DevOps/node-docker-app$ docker build -t node-docker-app .
[+] Building 0.0s (1/1) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile             0.0s
=> => transferring dockerfile: 2B                               0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory
ish@meow:~/DevOps/node-docker-app$ nano Dockerfile
ish@meow:~/DevOps/node-docker-app$ docker build -t node-docker-app .
[+] Building 41.4s (10/10) FINISHED                             docker:default
=> [internal] load build definition from Dockerfile             0.0s
=> => transferring dockerfile: 170B                             0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine 15.3s
=> [internal] load .dockerignore                               0.0s
=> => transferring context: 2B                                    0.0s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd 23.4s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3 0.0s
=> => sha256:ee77c6cd7c1886ecc802ad6cedef3a8ec1ea27d1fb9 6.18kB / 6.18kB 0.0s
=> => sha256:f18232174bc91741fdf3da96d85011092101a032a93 3.64MB / 3.64MB 4.3s
=> => sha256:dd71dde834b5c203d162902e6b8994cb2309ae04 40.01MB / 40.01MB 22.2s
=> => sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda0183 1.26MB / 1.26MB 6.0s
=> => sha256:8d6421d663b4c28fd3ebc498332f249011d11894558 7.67kB / 7.67kB 0.0s
=> => sha256:929b04d7c782f04f615cf785488fed452b6569f87c7 1.72kB / 1.72kB 0.0s
=> => extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b 0.1s
=> => sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9 446B / 446B 5.5s
=> => extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4 0.9s
=> => extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430 0.1s
=> => extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9e 0.0s
=> [internal] load build context                                0.1s
=> => transferring context: 2.70MB                                0.1s
=> [2/5] WORKDIR /app                                          0.1s
=> [3/5] COPY package*.json ./                                0.0s
=> [4/5] RUN npm install --production                          2.0s
=> [5/5] COPY . .                                              0.2s
=> exporting to image                                          0.2s
=> => exporting layers                                          0.2s
=> => writing image sha256:1aa96b8beb4b4498cc631a33fd38b1471d99849ccfa0b 0.0s
=> => naming to docker.io/library/node-docker-app             0.0s
ish@meow:~/DevOps/node-docker-app$
```

**Step5:** check whether the source code is pushed to git or not



-- → open the repo and refresh it now u can the source code as given in the below image

